

1. On the first line, we created a new instance of the array bannerImg, and gave it three data elements. (Remember, we are only making a copy of the Array object here.)
2. Next, we created two variables: newBanner, which has a beginning value of zero; and totalBan, which returns the length of the array (the total number of elements contained in the array).
3. Then we created a function named cycleBan. This function will be used to create a loop to cycle the images.
 - a. We set the newBanner variable to be increased each time the function cycles. *(Review: By placing the increment operator [" ++ "] after the variable [the "operand"], the variable is incremented only after it returns its current value to the script. For example, its beginning value is "0", so in the first cycle it will return a value of "0" to the script and then its value will be increased by "1".)*
 - b. When the value of the newBanner variable is equal to the variable totalBan (which is the length of the array), it is then reset to "0". This allows the images to start the cycle again, from the beginning.
 - c. The next statement uses the Document Object Method (DOM - we'll be taking a look at that soon) to display the images on the Web page. Remember, we use the dot operator to access the properties of an object. We also read the statement backwards, i.e., "take the element from the array bannerImg, that is specified by the current value of the variable newBanner, and place it in the src attribute located in the element with the name attribute of banner, which is located in the document object."
 - d. We then used the setTimeout function to tell the script how long to display each image. This is always measured in milliseconds so, in this case, the function cycleBan is called every 3,000 milliseconds (i.e., every 3 seconds).
4. Finally, we used the window.onload statement to execute the function cycleBan as soon as the document is loaded.

There are a total of five properties for the Array object. In addition to the length property listed above, the others are:

1. *constructor*: Specifies the function that creates an object's prototype.
2. *index*: Only applies to JavaScript arrays created by a regular expression match.
3. *input*: Only applies to JavaScript arrays created by a regular expression match.
4. *prototype*: Used to add properties or methods.

JAVASCRIPT Notes

The other properties listed here are either more advanced or seldom used. For now, we'll stick to the basics.

Javascript Object Hierarchy

Hierarchy Objects			
Object	Properties	Methods	Event Handlers
Window	defaultStatus frames opener parent scroll self status top window	alert blur close confirm focus open prompt clearTimeout setTimeout	onLoad onUnload onBlur onFocus
History	length forward go	back	none
Navigator	appName appVersion mimeTypes plugins userAgent	javaEnabled	none
document	alinkColor anchors applets area bgColor cookie fgColor forms images lastModified linkColor links location referrer title	clear close open write writeln	none (the onLoad and onUnload event handlers belong to the Window object.)

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	vlinkColor		
image	border complete height hspace lowsrc name src vspace width	none	none
form	action elements encoding FileUpload method name target	submit reset	onSubmit onReset
text	defaultValue name type value	focus blur select	onBlur onCharge onFocus onSelect

Built-in Objects

Array	length	join reverse sort xx	none
Date	none	getDate getDay getHours getMinutes getMonth getSeconds getTime getTimeZoneoffset getYear parse prototype setDate setHours setMinutes setMonth setSeconds setTime	none

		setYear toGMTString toLocaleString UTC	
String	length prototype	anchor big blink bold charAt fixed fontColor fontSize indexOf italics lastIndexOf link small split strike sub substring sup toLowerCase toUpperCase	Window

JavaScript Array Object

The Array object is used to store multiple values in a single variable.

Create an Array

The following code creates an Array object called myCars:

```
var myCars=new Array();
```

There are two ways of adding values to an array (you can add as many values as you need to define as many variables you require).

1:

```
var myCars=new Array();  
myCars[0]="Saab";  
myCars[1]="Volvo";  
myCars[2]="BMW";
```

JAVASCRIPT Notes

You could also pass an integer argument to control the array's size:

```
var myCars=new Array(3);  
myCars[0]="Saab";  
myCars[1]="Volvo";  
myCars[2]="BMW";
```

2:

```
var myCars=new Array("Saab","Volvo","BMW");
```

Note: If you specify numbers or true/false values inside the array then the type of variables will be numeric or Boolean instead of string.

Access an Array

You can refer to a particular element in an array by referring to the name of the array and the index number. The index number starts at 0.

The following code line:

```
document.write(myCars[0]);
```

will result in the following output:

```
Saab
```

Modify Values in an Array

To modify a value in an existing array, just add a new value to the array with a specified index number:

```
myCars[0]="Opel";
```

Now, the following code line:

```
document.write(myCars[0]);
```

will result in the following output:

```
Opel
```

JavaScript Date Object

Create a Date Object

The Date object is used to work with dates and times.

The following code create a Date object called myDate:

```
var myDate=new Date()
```

Note: The Date object will automatically hold the current date and time as its initial value!

Set Dates

We can easily manipulate the date by using the methods available for the Date object.

In the example below we set a Date object to a specific date (14th January 2010):

```
var myDate=new Date();  
myDate.setFullYear(2010,0,14);
```

And in the following example we set a Date object to be 5 days into the future:

```
var myDate=new Date();  
myDate.setDate(myDate.getDate()+5);
```

Note: If adding five days to a date shifts the month or year, the changes are handled automatically by the Date object itself!

Compare Two Dates

The Date object is also used to compare two dates.

The following example compares today's date with the 14th January 2010:

```
var myDate=new Date();  
myDate.setFullYear(2010,0,14);  
var today = new Date();  
if (myDate>today)  
{  
alert("Today is before 14th January 2010");  
}  
else  
{  
alert("Today is after 14th January 2010");  
}
```

JavaScript Math Object

Math Object

The Math object allows you to perform mathematical tasks.

The Math object includes several mathematical constants and methods.

Syntax for using properties/methods of Math:

```
var pi_value=Math.PI;  
var sqrt_value=Math.sqrt(16);
```

Note: Math is not a constructor. All properties and methods of Math can be called by using Math as an object without creating it.

Mathematical Constants

JavaScript provides eight mathematical constants that can be accessed from the Math object. These are: E, PI, square root of 2, square root of 1/2, natural log of 2, natural log of 10, base-2 log of E, and base-10 log of E.

You may reference these constants from your JavaScript like this:

```
Math.E  
Math.PI  
Math.SQRT2  
Math.SQRT1_2  
Math.LN2  
Math.LN10  
Math.LOG2E  
Math.LOG10E
```

Mathematical Methods

In addition to the mathematical constants that can be accessed from the Math object there are also several methods available.

The following example uses the round() method of the Math object to round a number to the nearest integer:

```
document.write(Math.round(4.7));
```

The code above will result in the following output:

```
5
```

JAVASCRIPT Notes

The following example uses the random() method of the Math object to return a random number between 0 and 1:

```
document.write(Math.random());
```

The code above can result in the following output:

```
0.4218824567728053
```

The following example uses the floor() and random() methods of the Math object to return a random number between 0 and 10:

```
document.write(Math.floor(Math.random()*11));
```

The code above can result in the following output:

```
4
```

JavaScript String Object

String object

The String object is used to manipulate a stored piece of text.

Examples of use:

The following example uses the length property of the String object to find the length of a string:

```
var txt="Hello world!";  
document.write(txt.length);
```

The code above will result in the following output:

```
12
```

The following example uses the toUpperCase() method of the String object to convert a string to uppercase letters:

```
var txt="Hello world!";  
document.write(txt.toUpperCase());
```

The code above will result in the following output:

```
HELLO WORLD!
```


Window Object

The Window object is the top level object in the JavaScript hierarchy.

The Window object represents a browser window.

A Window object is created automatically with every instance of a <body> or <frameset> tag.

IE: Internet Explorer, **F:** Firefox, **O:** Opera.

Window Object Collections

Collection	Description	IE	F	O
frames[]	Returns all named frames in the window	4	1	9

Window Object Properties

Property	Description	IE	F	O
closed	Returns whether or not a window has been closed	4	1	9
defaultStatus	Sets or returns the default text in the statusbar of the window	4	No	9
document	See Document object	4	1	9
history	See History object	4	1	9
length	Sets or returns the number of frames in the window	4	1	9
location	See Location object	4	1	9
name	Sets or returns the name of the window	4	1	9
opener	Returns a reference to the window that created the window	4	1	9
outerHeight	Sets or returns the outer height of a window	No	1	No
outerWidth	Sets or returns the outer width of a window	No	1	No
pageXOffset	Sets or returns the X position of the current page in relation to the upper left corner of a window's display area	No	No	No
pageYOffset	Sets or returns the Y position of the current page in relation to the upper left corner of a window's display area	No	No	No
parent	Returns the parent window	4	1	9
personalbar	Sets whether or not the browser's personal bar (or directories bar) should be visible			
scrollbars	Sets whether or not the scrollbars should be visible			
self	Returns a reference to the current window	4	1	9
status	Sets the text in the statusbar of a window	4	No	9
statusbar	Sets whether or not the browser's statusbar should be visible			
toolbar	Sets whether or not the browser's tool bar is visible or not (can only be set before the window is opened and you must have UniversalBrowserWrite privilege)			
top	Returns the topmost ancestor window	4	1	9

JAVASCRIPT Notes

Window Object Methods

Method	Description	IE	F	O
alert()	Displays an alert box with a message and an OK button	4	1	9
blur()	Removes focus from the current window	4	1	9
clearInterval()	Cancels a timeout set with setInterval()	4	1	9
clearTimeout()	Cancels a timeout set with setTimeout()	4	1	9
close()	Closes the current window	4	1	9
confirm()	Displays a dialog box with a message and an OK and a Cancel button	4	1	9
createPopup()	Creates a pop-up window	4	No	No
focus()	Sets focus to the current window	4	1	9
moveBy()	Moves a window relative to its current position	4	1	9
moveTo()	Moves a window to the specified position	4	1	9
open()	Opens a new browser window	4	1	9
print()	Prints the contents of the current window	5	1	9
prompt()	Displays a dialog box that prompts the user for input	4	1	9
resizeBy()	Resizes a window by the specified pixels	4	1	9
resizeTo()	Resizes a window to the specified width and height	4	1.5	9
scrollBy()	Scrolls the content by the specified number of pixels	4	1	9
scrollTo()	Scrolls the content to the specified coordinates	4	1	9
setInterval()	Evaluates an expression at specified intervals	4	1	9
setTimeout()	Evaluates an expression after a specified number of milliseconds	4	1	9

Document Object

The Document object represents the entire HTML document and can be used to access all elements in a page.

The Document object is part of the Window object and is accessed through the window.document property.

IE: Internet Explorer, **F:** Firefox, **O:** Opera, **W3C:** World Wide Web Consortium (Internet Standard).

Document Object Collections

Collection	Description	IE	F	O	W3C
anchors[]	Returns a reference to all Anchor objects in the document	4	1	9	Yes
forms[]	Returns a reference to all Form objects in the document	4	1	9	Yes
images[]	Returns a reference to all Image objects in the document	4	1	9	Yes
links[]	Returns a reference to all Area and Link objects in	4	1	9	Yes

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	the document				
--	--------------	--	--	--	--

Document Object Properties

Property	Description	IE	F	O	W3C
body	Gives direct access to the <body> element				
cookie	Sets or returns all cookies associated with the current document	4	1	9	Yes
domain	Returns the domain name for the current document	4	1	9	Yes
lastModified	Returns the date and time a document was last modified	4	1	No	No
referrer	Returns the URL of the document that loaded the current document	4	1	9	Yes
title	Returns the title of the current document	4	1	9	Yes
URL	Returns the URL of the current document	4	1	9	Yes

Document Object Methods

Method	Description	IE	F	O	W3C
close()	Closes an output stream opened with the document.open() method, and displays the collected data	4	1	9	Yes
getElementById()	Returns a reference to the first object with the specified id	5	1	9	Yes
getElementsByName()	Returns a collection of objects with the specified name	5	1	9	Yes
getElementsByTagName()	Returns a collection of objects with the specified tagname	5	1	9	Yes
open()	Opens a stream to collect the output from any document.write() or document.writeln() methods	4	1	9	Yes
write()	Writes HTML expressions or JavaScript code to a document	4	1	9	Yes
writeln()	Identical to the write() method, with the addition of writing a new line character after each expression	4	1	9	Yes

History Object

The History object is actually a JavaScript object, not an HTML DOM object.

The History object is automatically created by the JavaScript runtime engine and consists of an array of URLs. These URLs are the URLs the user has visited within a browser window.

The History object is part of the Window object and is accessed through the window.history property.

IE: Internet Explorer, **F:** Firefox, **O:** Opera.

JAVASCRIPT Notes

History Object Properties

Property	Description	IE	F	O
length	Returns the number of elements in the history list	4	1	9

History Object Methods

Method	Description	IE	F	O
back()	Loads the previous URL in the history list	4	1	9
forward()	Loads the next URL in the history list	4	1	9
go()	Loads a specific page in the history list	4	1	9

Form Object

The Form object represents an HTML form.

For each instance of a <form> tag in an HTML document, a Form object is created.

IE: Internet Explorer, **F:** Firefox, **O:** Opera, **W3C:** World Wide Web Consortium (Internet Standard).

Form Object Collections

Collection	Description	IE	F	O	W3C
elements[]	Returns an array containing each element in the form	5	1	9	Yes

Form Object Properties

Property	Description	IE	F	O	W3C
acceptCharset	Sets or returns a list of possible character-sets for the form data	No	No	No	Yes
action	Sets or returns the action attribute of a form	5	1	9	Yes
enctype	Sets or returns the MIME type used to encode the content of a form	6	1	9	Yes
id	Sets or returns the id of a form	5	1	9	Yes
length	Returns the number of elements in a form	5	1	9	Yes
method	Sets or returns the HTTP method for sending data to the server	5	1	9	Yes
name	Sets or returns the name of a form	5	1	9	Yes
target	Sets or returns where to open the action-URL in a form	5	1	9	Yes

Standard Properties

Property	Description	IE	F	O	W3C
className	Sets or returns the class attribute of an element	5	1	9	Yes
dir	Sets or returns the direction of text	5	1	9	Yes
lang	Sets or returns the language code for an element	5	1	9	Yes
title	Sets or returns an element's advisory title	5	1	9	Yes

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Form Object Methods

Method	Description	IE	F	O	W3C
reset()	Resets the values of all elements in a form	5	1	9	Yes
submit()	Submits a form	5	1	9	Yes

Image Object

The Image object represents an embedded image.

For each instance of an tag in an HTML document, an Image object is created.

IE: Internet Explorer, **F:** Firefox, **O:** Opera, **W3C:** World Wide Web Consortium (Internet Standard).

Image Object Properties

Property	Description	IE	F	O	W3C
align	Sets or returns how to align an image according to the surrounding text	5	1	9	Yes
alt	Sets or returns an alternate text to be displayed, if a browser cannot show an image	5	1	9	Yes
border	Sets or returns the border around an image	4	1	9	Yes
complete	Returns whether or not the browser has finished loading the image	4	1	9	No
height	Sets or returns the height of an image	4	1	9	Yes
hspace	Sets or returns the white space on the left and right side of the image	4	1	9	Yes
id	Sets or returns the id of the image	4	1	9	Yes
isMap	Returns whether or not an image is a server-side image map	5	1	9	Yes
longDesc	Sets or returns a URL to a document containing a description of the image	6	1	9	Yes
lowsrc	Sets or returns a URL to a low-resolution version of an image	4	1	9	No
name	Sets or returns the name of an image	4	1	9	Yes
src	Sets or returns the URL of an image	4	1	9	Yes
useMap	Sets or returns the value of the usemap attribute of an client-side image map	5	1	9	Yes
vspace	Sets or returns the white space on the top and bottom of the image	4	1	9	Yes
width	Sets or returns the width of an image	4	1	9	Yes

Standard Properties

Property	Description	IE	F	O	W3C
className	Sets or returns the class attribute of an element	5	1	9	Yes
title	Sets or returns an element's advisory title	5	1	9	Yes

Area Object

The Area object represents an area of an image-map (An image-map is an image with clickable regions).

For each instance of an <area> tag in an HTML document, an Area object is created.

IE: Internet Explorer, **F:** Firefox, **O:** Opera, **W3C:** World Wide Web Consortium (Internet Standard).

Area Object Properties

Property	Description	IE	F	O	W3C
accessKey	Sets or returns the keyboard key to access an area	5	1	No	Yes
alt	Sets or returns an alternate text to be displayed, if a browser cannot show an area	5	1	9	Yes
coords	Sets or returns the coordinates of a clickable area in an image-map	5	1	9	Yes
hash	Sets or returns the anchor part of the URL in an area	4	1	No	No
host	Sets or returns the hostname and port of the URL in an area	4	1	No	No
href	Sets or returns the URL of a link in an image-map	4	1	9	Yes
id	Sets or returns the id of an area	4	1	9	Yes
noHref	Sets or returns whether an area should be active or inactive	5	1	9	Yes
pathname	Sets or returns the pathname of the URL in an area	4	1	9	No
protocol	Sets or returns the protocol of the URL in an area	4	1	9	No
search	Sets or returns the query string part of the URL in an area	4	1	9	No
shape	Sets or returns the shape of an area in an image-map	5	1	9	Yes
tabIndex	Sets or returns the tab order for an area	5	1	9	Yes
target	Sets or returns where to open the link-URL in an area	4	1	9	Yes

Standard Properties

Property	Description	IE	F	O	W3C
className	Sets or returns the class attribute of an element	5	1	9	Yes
dir	Sets or returns the direction of text	5	1	9	Yes
lang	Sets or returns the language code for an element	5	1	9	Yes
title	Sets or returns an element's advisory title	5	1	9	Yes

Navigator Object

The Navigator object is actually a JavaScript object, not an HTML DOM object.

The Navigator object is automatically created by the JavaScript runtime engine and contains information about the client browser.

IE: Internet Explorer, **F:** Firefox, **O:** Opera.

JAVASCRIPT Notes

Navigator Object Collections

Collection	Description	IE	F	O
plugins[]	Returns a reference to all embedded objects in the document	4	1	9

Navigator Object Properties

Property	Description	IE	F	O
appName	Returns the code name of the browser	4	1	9
appMinorVersion	Returns the minor version of the browser	4	No	No
appName	Returns the name of the browser	4	1	9
appVersion	Returns the platform and version of the browser	4	1	9
browserLanguage	Returns the current browser language	4	No	9
cookieEnabled	Returns a Boolean value that specifies whether cookies are enabled in the browser	4	1	9
cpuClass	Returns the CPU class of the browser's system	4	No	No
onLine	Returns a Boolean value that specifies whether the system is in offline mode	4	No	No
platform	Returns the operating system platform	4	1	9
systemLanguage	Returns the default language used by the OS	4	No	No
userAgent	Returns the value of the user-agent header sent by the client to the server	4	1	9
userLanguage	Returns the OS' natural language setting	4	No	9

Navigator Object Methods

Method	Description	IE	F	O
javaEnabled()	Specifies whether or not the browser has Java enabled	4	1	9
taintEnabled()	Specifies whether or not the browser has data tainting enabled	4	1	9

ZIP CODE VALIDATION

<!-- TWO STEPS TO INSTALL ZIP CODE VALIDATION:

1. Copy the coding into the HEAD of your HTML document
2. Add the last code into the BODY of your HTML document -->

<!-- STEP ONE: Paste this code into the HEAD of your HTML document -->

<HEAD>

<SCRIPT LANGUAGE="JavaScript">

<!-- Original: Brian Swalwell -->

<!-- This script and many more are available free online at -->

<!-- The JavaScript Source!! <http://javascript.internet.com> -->

<!-- Begin

```
function validateZIP(field) {
```

```
var valid = "0123456789-";
```

```
var hyphencount = 0;
```

```
if (field.length!=5 && field.length!=10) {
```

```
alert("Please enter your 5 digit or 5 digit+4 zip code.");
```

```
return false;
```

```
}
```

```
for (var i=0; i < field.length; i++) {
```

```
temp = "" + field.substring(i, i+1);
```

```
if (temp == "-") hyphencount++;
```

```
if (valid.indexOf(temp) == "-1") {
```

```
alert("Invalid characters in your zip code. Please try again.");
```

```
return false;
```

```
}
```

```
if ((hyphencount > 1) || ((field.length==10) && ""+field.charAt(5)!="-")) {
```

```
alert("The hyphen character should be used with a properly formatted 5 digit+four zip code, like '12345-6789'. Please try again.");
```

```
return false;
```

```
}
```

```
}
```

```
return true;
```



```
}
```

```
// End -->
```

```
</script>
```

```
</HEAD>
```

```
<!-- STEP TWO: Copy this code into the BODY of your HTML document -->
```

```
<BODY>
```

```
<center>
```

```
<form name=zip onSubmit="return validateZIP(this.zip.value)">
```

```
Zip: <input type=text size=30 name=zip>
```

```
<input type=submit value="Submit">
```

```
</form>
```

```
</center>
```

```
<p><center>
```

```
<font face="arial, helvetica" size="-2">Free JavaScripts provided<br>
```

```
by <a href="http://javascriptsource.com">The JavaScript Source</a></font>
```

```
</center><p>
```